Alex Morgan

San Francisco, CA | +1 (555) 019-2834 | alex.morgan@example.com | linkedin.com/in/alexmorgan | github.com/alexmorgan

Education

University of California, Berkeley — Berkeley, CA
*Bachelor of Science in Electrical Engineering & Computer Sciences – **GPA: 3.92/4.0*** — *Aug 2022 – May 2026*

Technical Skills

Languages: C, C++, Python, Rust, SQL, Bash, Go
Embedded & Hardware: ARM Cortex-M, STM32, ESP32, FreeRTOS, Embedded Linux, PCB Design (KiCad), I2C, SPI, UART, CAN
Robotics & Systems: ROS/ROS 2, Gazebo, Kalman Filtering, SLAM, PID Control, OpenCV, Linux Kernel Drivers
Protocols & Cloud: MQTT, Modbus, TCP/IP, BLE, LoRa, AWS IoT Core, Docker, Git, CI/CD Actions

Experience

Robotics Innovations Inc. — San Jose, CA
Embedded Software Engineering Intern — *May 2025 – Aug 2025*

– Developed real-time motor control and sensor fusion firmware on **STM32F4** microcontrollers using **FreeRTOS**.
– Engineered high-speed **CAN bus** telemetry communication pipeline achieving <5ms end-to-end sensor transmission latency.
– Integrated automated Hardware-in-the-Loop (HIL) testing in GitHub Actions, decreasing regression debugging time by 35%.

Autonomous Systems Lab, UC Berkeley — Berkeley, CA
Undergraduate Research Assistant — *Jan 2024 – May 2025*

– Designed and built a distributed sensor payload for unmanned aerial vehicles (UAVs) using **ESP32** and **LoRa** radios.
– Authored custom Linux device drivers for I2C environmental telemetry sensors with real-time DMA transfers.

Key Projects

Autonomous Pipeline Inspection Rover — Jan 2025 – Present
Technologies: C++, ROS 2, FreeRTOS, ESP32-S3, OpenCV, RS485

– Designed a compact tracked robot for industrial pipeline inspection with real-time visual SLAM and obstacle avoidance.

High-Precision Industrial IoT Data Logger — Aug 2024 – Dec 2024
Technologies: C++, FreeRTOS, Modbus RTU, MQTT, AWS IoT, SD Card Flash

– Built an ultra-low-power industrial data logger supporting Modbus telemetry with local flash caching and zero data loss on network dropouts.

Publications & Leadership

- **IEEE Conference Publication:** *"Sensor Fusion for Autonomous Ground Navigation in Constrained Environments"*, IEEE IROS Workshop (2025).
- **Leadership:** President, IEEE Student Branch at UC Berkeley (Managed 20+ technical workshops and 300+ members).

Honors & Certifications

Dean's Honors List: UC Berkeley (4 Semesters)
Embedded Systems Certification: Arm Education (Score: 95%)
1st Place Winner: California Collegiate Robotics Hackathon 2024
AWS Certified: Solutions Architect Associate